

Climbing Safety System Design in Mount Prau using VHF Radio Signal

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Abstract. An empirical experiment was conducted on the climbing safety system on Mount Prau using VHF radio signals. In the experiments carried out, the data obtained in the form of position and readability were then processed to obtain data on the distribution and signal strength along the hiking trail. From the research that has been done, the design of a climbing safety system needs to be made to make it safe and comfortable for mountain climbers. The main climbing safety system includes speed in handling accidents. To support this, a communication tool is needed which in this research was tested in the signal aspect. This research also revealed that the VHF radio signal along the Mount Prau hiking trail has good quality with an average signal strength of 41.3226 V/m. With the results of this research, it is possible to carry out further development to create a climbing safety system in the form of a fixed radio station, especially at each climbing post.

Keywords: *climbing, mountaineering, radio communication, safety system design, communication network*

INTRODUCTION

According to a survey held by Statista Research Department, the number of climbers in 2019 has risen significantly by nearly 20 million, which is 66.44% from 2006. This has a positive impact on the community because mountain climbing is a healthy activity, but it demands good physical form and health, so it requires good preparation [1]. Unfortunately, there's a lot of people that do not have the basic knowledge and skills in mountain climbing. As a result, many climbers got lost and many accidents occurred. Considering that in the last five years, 80% of the population had not taken any applicable courses, such as basic first aid courses [1]. There are 30% of deaths that are related to mountain activities in France are mountain climbing, with 26 as the average number of victims per year from 2003 to 2012 [2].

Based on the accident cases that happened, it can be seen that one of the preventive measures is quick first aid response. When an accident occurs in a hiking trail, climbers who do not have the knowledge of first aid need help from others, so they need to report the accident to the mountain administrator in the base camp which takes a lot of time for the first aid and/or advanced help to arrive. Therefore, a communication network is needed so that the time required to reach the base camp can be shortened so the climbers don't need to go down to report an emergency or ask for help.

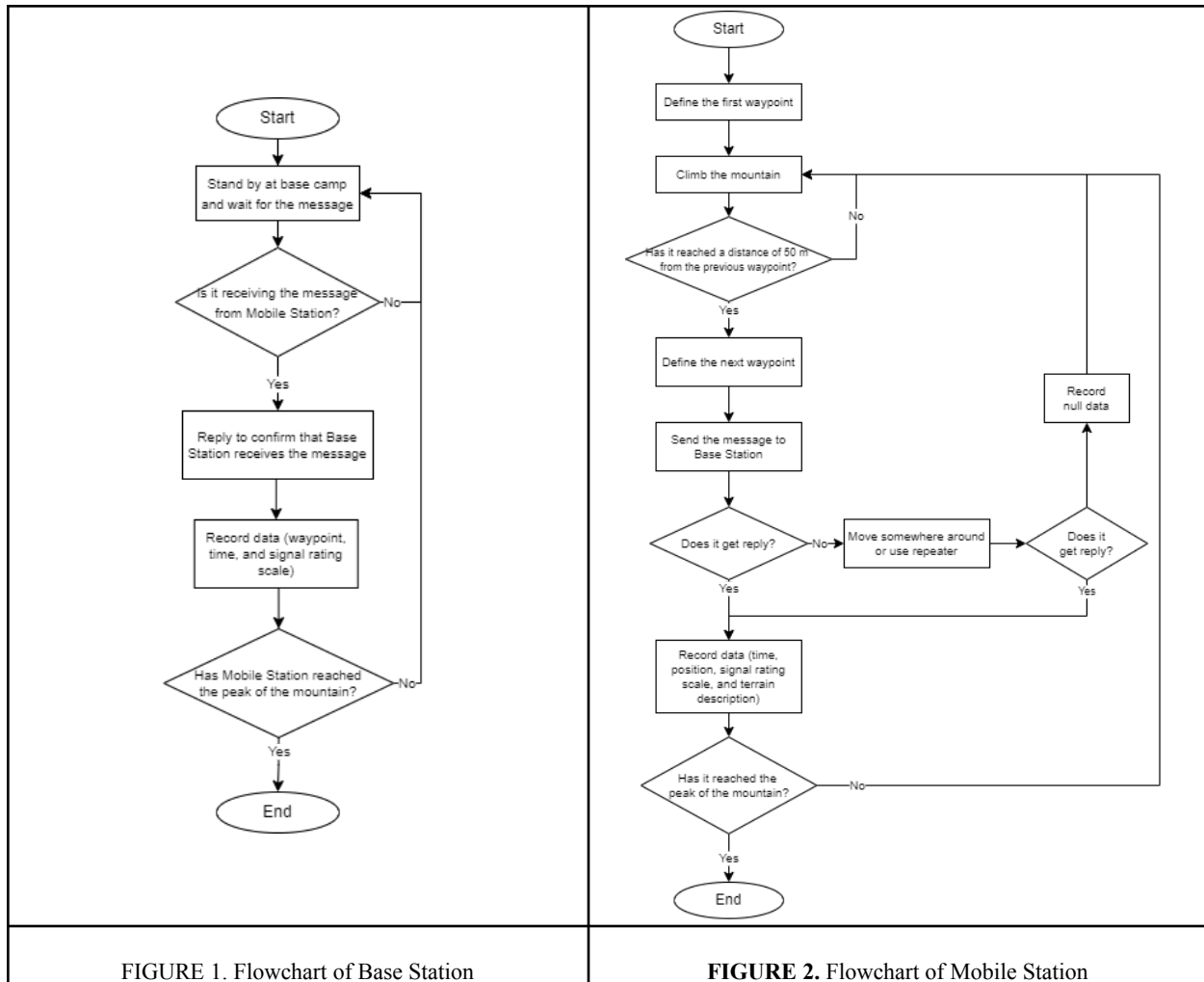
The communication which will be used is radio communications. For radio communications to occur, there must be a transmitter that takes sound or data, converts it into a radio frequency (RF) signal, and transmits it through an antenna. There must also be a receiver that receives the RF signal and converts it back into sound or data. Radios that contain both a transmitter and a receiver are referred to as transceivers [3]. The RF which will be used in communication networks is Very High Frequency (VHF). The wavelength of VHF is between 30 MHz - 300 MHz and 150 km range. The advantages of using radio frequency communication in this situation are that it is easier and faster to implement, more affordable, and has a wide range.

Handheld-Transceiver (HT) will be used for this research to know how well radio signals behave in Prau Mountain so it can see if this communication can be implemented or not. Afterward, a safety system will be designed based on the research result. Other than that, if this system can be implemented, that system can also function as a tool for the mountain ranger in communicating with mountain administrators for other reasons not only for emergency conditions. This system hopefully could support local governments to promote mountain tourism while giving more sense of comfort to the climbers by reducing the number of accidents with an available communication network for emergency use.

METHOD

Ideal climbing conditions have a high level of safety and an adequate safety system, where there is a system that connects climbers, base camp, and rescue teams. To achieve this condition, the research was conducted on the safety system on Mount Prau, by looking at it from a communication point of view. In this research an empirical method is used, this method is implemented by signal testing from the base camp to the peak of Mount Prau. The tools used are a pair of handheld-transceivers (HT), one of which uses a Larsen antenna and the other uses a super stick antenna, and a global positioning system (GPS). Global positioning system (GPS) is a space-based radio navigation system owned by the United States government and operated by the United States Air Force. It is a global navigation satellite system that provides geolocation and time information to a GPS receiver anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites [4]. The method that will be used is to communicate directly between HT and HT by using a frequency of 153.30 MHz which is the frequency used by the Mount Prau administrator.

There is an alternative method of using a repeater that is already installed on the mountain's peak, but it's used by the local amateur radio satellite organization (ORARI) and is not available for the general public if there is no permit. There is also an alternative additional method of sending simple coordination of Global Satellite Radio Navigation Systems that could help to give a location, but it has a limited accuracy of 19 meters error using NAVSTAR SRNS and 14 meters error using SRNS GLONASS SYSTEM. Those errors are deemed too big of an error to be added as a feature. [5]



Based on Figure 1 a handheld transceiver (HT) was prepared which was using a Larsen antenna placed at the Base Station will stand by at the base camp and wait for messages from Mobile Station. If there is a message, then Base Station will confirm that the message was received. Data such as waypoint, time, and signal rating scale were recorded based on the message that was received. This is done repeatedly until the Mobile Station reaches the peak of the mountain. Meanwhile, in Figure 2, an HT which was using a super stick antenna is carried when climbing from the base camp to the peak of the mountain. The mobile station defines the first waypoint and then climbs the mountain trail until reaches a 50-meter distance using GPS. This method continues until it reaches the peak of the mountain. When there is no reply from the base station, the mobile station has to move somewhere around or use the second method which is using a repeater. If there is still no reply, then still need to record data that there is no reply or null data.

Both stations use HT to send messages alternately to avoid stacking. If there is a process of transmitting from two or more communication devices, it will cause a buildup or jump of signals on one frequency which results in the closing of audio information in communication [6]. GPS is used to determine location coordinates and determine the distance traveled when tracking. Prior to tracking, the coordinates of the place, the altitude of the base camp, and the rating scale of messages sent between the mobile station and the base station are recorded. Testing the rating scale of this message is done by sending a formatted message. This message is sent from the mobile station HT to the base station HT. The message rating scale is recorded in the form of a quantity value from 1 to 5, based on the signal rating scale based on the overall rating on the SINPO Codes from the International Telecommunications Union-Radiocommunication Sector (ITU-R). SINPO is an acronym for several parameters that describe how well the signal is used to convey messages through radio signals. However, in this research, only an overall rating was

used with a value of 1 being the lowest signal rating scale and a value of 5 being the highest signal rating scale, as can be seen in Table 1.

TABLE 1. SINPO signal reporting code [7]					
Rating Scale	S	I	N	P	O
	Signal strength	Degrading effect of			Overall rating
		Interference	Noise	Propagation disturbance	
5	Excellent	Nil	Nil	Nil	Excellent
4	Good	Slight	Slight	Slight	Good
3	Fair	Moderate	Moderate	Moderate	Fair
2	Poor	Severe	Severe	Severe	Poor
1	Barely audible	Extreme	Extreme	Extreme	Unusable

Coordinate data and altitude data that have been obtained are processed into distance data that shows the distance between the mobile station that sends the message and the base station that confirms the message. On the mobile station, a description of the surrounding terrain has been recorded. However, because almost all terrain tends to be homogeneous, so the terrain description is negligible.

RESULT AND DISCUSSION

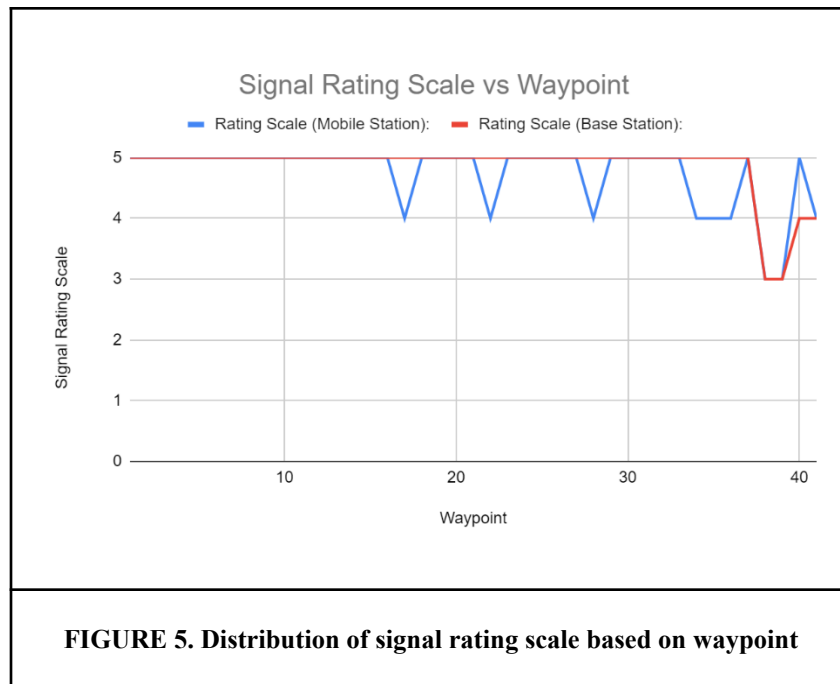
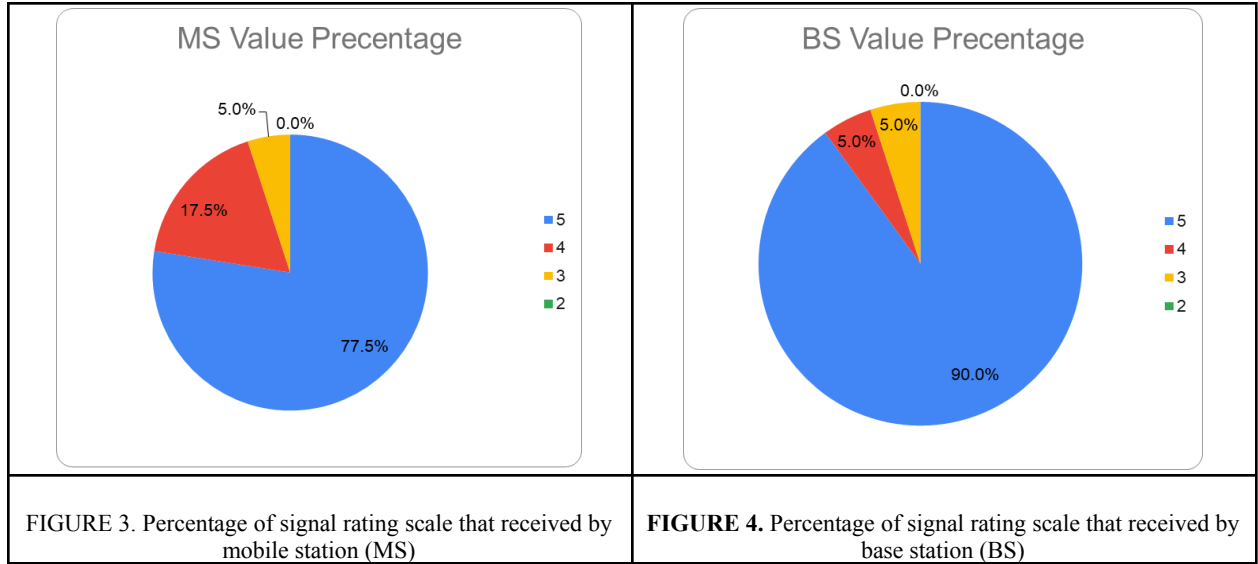
The empirical method is used to obtain position data and rating scale in this research. Position data is obtained in the form of latitude, longitude, and altitude. In addition, signal rating scale data is also obtained. The signal rating scale data is divided into two parts, namely the rating scale of the mobile station and the rating scale of the base station. The mobile station scale rating is the rating scale received by the mobile station and the base station scale rating is the rating scale received by the base station. All data are recorded and displayed in Table 2.

TABLE 2. Data of Signal Rating Scale, Coordinates, Elevation, and Distance Based on Research

No	Waypoint	Position			Rating Scale		
		Latitude	Longitude	Altitude	Distance from Base camp (m)	Mobile Station	Base Station
1	T1	-7.20949	109.9253	2010.7	0	5	5
2	T2	-7.20909	109.9251	2007.62	49.65	5	5
3	T3	-7.20885	109.9246	2007.79	105.01	5	5
4	T4	-7.2083	109.9246	2008.28	153.21	5	5
5	T5	-7.20798	109.9249	2016.8	173.61	5	5
6	T6	-7.20754	109.9251	2016.72	217.95	5	5
7	T7	-7.20718	109.9255	2016.89	257.81	5	5
8	T8	-7.20695	109.9258	2030.89	287.77	5	5
9	T9	-7.20727	109.9262	2040.43	266.07	5	5
10	T10	-7.20755	109.9264	2043.88	247.51	5	5
11	T11	-7.20805	109.9267	2049.29	222.47	5	5
12	T12	-7.20849	109.9268	2060.57	199.36	5	5
13	T13	-7.2089	109.927	2062.72	198.68	5	5
14	T14	-7.20908	109.9275	2063.9	246.94	5	5

1	T15						
5		-7.20932	109.9279	2069.42	287.44	5	5
1	T16						
6		-7.20945	109.9283	2075.63	330.98	5	5
1	T17						
7		-7.20945	109.9288	2081.7	386.13	4	5
1	T18						
8		-7.20957	109.9294	2085.11	452.38	5	5
1	T19						
9		-7.20907	109.9291	2131.06	421.79	5	5
2	T20						
0		-7.20866	109.929	2132.85	418.47	5	5
2	T21						
1		-7.20806	109.929	2137.8	438.05	5	5
2	T22						
2		-7.20766	109.9289	2200.48	446.23	4	5
2	T23						
3		-7.20724	109.9291	2206.65	488.18	5	5
2	T24						
4		-7.20696	109.9294	2213.02	532.65	5	5
2	T25						
5		-7.20675	109.9301	2178.79	610.91	5	5
2	T26						
6		-7.20626	109.9305	2195.86	676.8	5	5
2	T27						
7		-7.20619	109.9309	2201.05	718.53	5	5
2	T28						
8		-7.20574	109.931	2304.5	754.5	4	5
2	T29						
9		-7.20532	109.9312	2311.14	799.14	5	5
3	T30						
0		-7.20484	109.9315	2315.08	857.41	5	5
3	T31						
1		-7.20431	109.9317	2324.78	911.17	5	5
3	T32						
2		-7.20408	109.932	2363.7	952.98	5	5
3	T33						
3		-7.20372	109.9324	2448.99	1012.48	5	5
3	T34						
4		-7.20338	109.9328	2446.24	1070.58	4	5
3	T35						
5		-7.20329	109.9333	2474.11	1119.89	4	5
3	T36						
6		-7.20322	109.9337	2497.84	1159.64	4	5
3	T37						
7		-7.20277	109.9342	2513.48	1233.82	5	5
3	T38						
8		-7.20267	109.9347	2566.62	1284.68	3	3
3	T39						
9		-7.20249	109.935	2569.4	1323.22	3	3
4	T40						
0		-7.20196	109.9353	2574.82	1384.94	5	4
4	T41						
1		-7.20105	109.9352	2577.65	1439.97	4	4

By using a method standardized by the International Telecommunication Union, the voice produced by the handheld transceiver of signal rating scale from one point to the base camp which is also the first point that is checked before the following 40 points before reaching the peak. Based on the overall SINPO score data that has been obtained, the percentage of excellent signal overall at the base station reaches 90% of the 40 points examined, 5% is good, and 5% is fair. While on the mobile station the signal rating scale reaches 77.5% excellent with 17.5% good and 5% fair. Both results are shown in Figure 3 and Figure 4. It can be seen that the signal from base camp to the top of Mount Prau has a very good rating scale with only 2 waypoints having a rating of 3 on the mobile station and base station while the rest have a rating above 3 which indicates that the creation of a radio communication system in Mount Prau is an applicable option.



With the latitude, longitude, and altitude contained in the data that has been presented, the distance between the point and the base camp point is sought. This can be solved using a simple Pythagorean by having the longitude and latitude values converted to meters as in Equation (1). This can be done using a conversion simplification at the value of the poles where 1 longitude and latitude can be converted to 111.32 km [8].

$$Distance = \sqrt{(Base\ Station\ Point\ (x, y, z) - Mobile\ Station\ Point(x, y, z))^2} \quad (1)$$

Based on Table 1 that shows the relationship between distance from base camp and signal rating scale, it can be seen that there is a decrease in signal rating scale corresponding to the distance between the mobile station and the base station, but the decline is still acceptable. In these conditions, the sound can still be deciphered. The signal

rating scale based on the quality of the voltage (E) can be quantized as microvolts per meter using the voltage gain formula which is written as in Equation (2) [9].

$$Gain = \frac{20 \log E}{50} \quad (2)$$

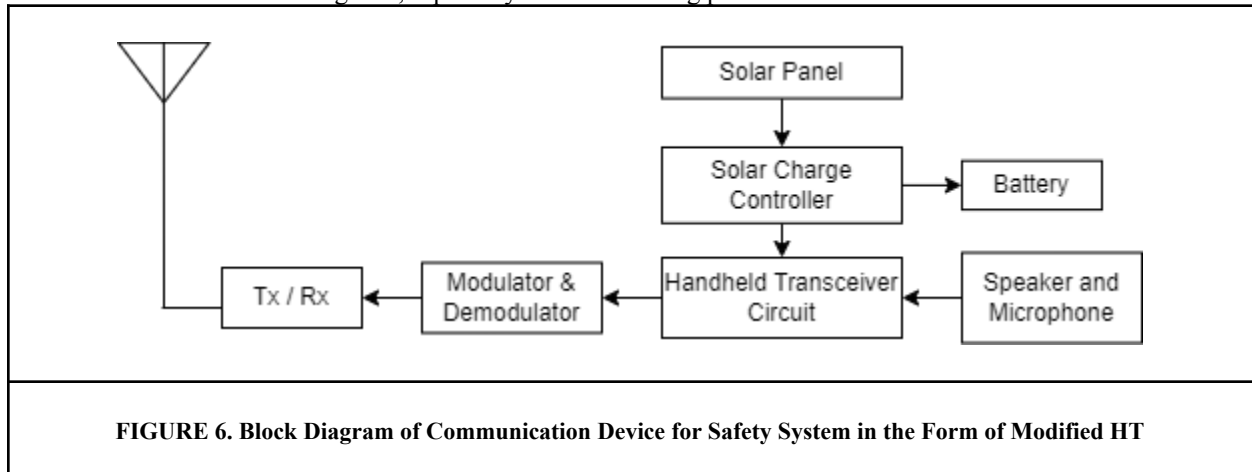
With the gain in decibels having a maximum value of 60, the overall value is converted from 5 to 60 dB which is a perfect signal, 4 to 48 dB, 3 to 36 dB, and so on. By using the above formula, the following formula is generated as in Equation (3).

$$E = 5 \times 10^{\frac{Gain}{20}} \quad (3)$$

$$Gain = Rating\ Scale \times 12\ dB \quad (4)$$

Input the required data in table 1 into Equation (4) and Equation (3), it will get an average of quality of the voltage (E) at the mobile station of 413222.58 $\mu V/m$ and at the base station of 45888.5 $\mu V/m$, it can be seen that the signal strength along the Mount Prau climbing trail has the potential to be used to support the mountain climbing safety system that is being tested. However, at some waypoints, it still experiences fluctuations, as can be seen from the signal rating scale which has decreased in Figure 5. Overall, from the experimental results that have been carried out and the results of data processing carried out, the signal strength along the tracking path is capable of supporting the climbing safety system design.

At the time of data collection, the weather on the climbing trail tends to be good with a little cloud and light rain. Factors such as climate are ignored in this research because altitude, humidity, and other factors depending on the region where the system is applied are homogeneous. From the results of the research, the application of a climbing safety system design for the Mount Prau climbing route has the capability to create a communication system to support the climbing safety system. With the results of signal distribution and signal strength, it is possible to build a fixed radio station on the hiking trail, especially at each climbing post.



Because this system is possible to make, the first thing that is proposed is the manufacture of a communication device for the safety system in the form of a modified HT. The modification that will be made is to make the HT fixed somewhere to be determined. For the electrical needs of the device, a solar panel electrical system will be used with a solar charge controller and a battery so that the device can be used all the time. In order to use energy efficiently, a directional antenna will be used so that no energy is wasted. In addition, modification of the shape of the HT needs to be done so that it has weather resistance and security from theft. The modifications made are adjusted as shown in the block diagram in Figure 6. Based on the results of the research, signal rating scale data show that the proposed device placement could be placed at all waypoints. However, for efficiency, it is proposed that the device should be placed at every climbing post where Post 1 is in between T18 and T19, Post 2 is at T25, Post 3 is at T33, and at the peak of the mountain, which is at T41. Table 1 is showing that all waypoint that will be proposed has at least 4 signal rating scale which indicates that the signal is good.

CONCLUSION

From the research that has been done, the use of radio frequencies along the Mount Prau hiking trail can be implemented with supporting tools such as modified handheld transceivers. Signal strength and distribution that have been mapped can be used as an opening for further research for the design of climbing safety systems, one of them being in the design and manufacture of fixed radio stations. That way, it is hoped that the climbing safety system can be implemented optimally, especially to increase the accident emergency.

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